

Day and Night: 24-Hour Control Room

GROW • Science • 40 minutes

I can use a rotating-Earth model to collect and explain evidence across one complete day-night cycle.

Prepare before pupils arrive

Setup: Prepare a clearly marked globe and a cool fixed lamp. Place a four-row recording sheet at each working space.

Safety: Secure cables, use an LED lamp that remains cool, preserve clear walking space and keep eyes away from direct glare.

Equipment: Cool LED lamp, marked globe or ball, four-row evidence sheet, arrows and pencil.

Safety: Do not stare into the lamp. Use a cool LED only, keep hands off plugs and keep movement clear of cables.

Fallback: Turn a marked paper Earth beside a fixed Sun card, or use four prepared diagrams and order them.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Explain day and night and the apparent movement of the Sun.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Last lesson: what moved in our model and what stayed fixed? Last lesson: why does the Sun appear to cross the sky?	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.
I do · 4 min	cycle — repeating sequence rotation — one full turn evidence — what we observed Place one marker on the globe and start it at sunrise, where it crosses into the lit half. Turn the globe west to east in equal quarter turns.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.

Stage / total time	Teaching move	Adult support / response
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Add an observer arrow at each marker position to show which way the local person faces. Describe the Sun's apparent east-to-west position without claiming that the lamp moved; west-to-east Earth rotation creates that observed direction.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Commit each control-room observation to the correct point in the cycle. Select one observation.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Run or represent one complete four-position rotation and produce a control-room evidence record. Record: Four ordered rows recording marker position, light state, apparent Sun position and one final because conclusion.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Use two evidence rows to explain one full day-night cycle. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move. The pupil writes what they said and what changed.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: The marker rotated onto the side facing away from the lamp.

Correct. The lamp stayed on; rotation moved the place from the lit half to the unlit half.

Misconception repair

Repaired. Clouds can reduce daylight, but local night occurs when rotation carries a place onto Earth's side facing away from the Sun.

Guided checkpoint

Correct. The lamp stayed on; rotation moved the place from the lit half to the unlit half. Use only evidence actually present in the model. What changed: the lamp, clouds or the marker's position?

Control row 1

Sunrise. Row 1 is sunrise: the place rotates into the lit half. Find the cycle point where local daylight begins.

Control row 2

Middle of the day. Row 2 is the middle of the day in this simplified model. Choose the point where the marker faces the lamp most directly.

Control row 3

Sunset. Row 3 is sunset: the place rotates out of the lit half. Find the point where local daylight ends.

Control row 4

Middle of the night. Row 4 is the middle of the night in this simplified model. Choose the point furthest from the lit-facing centre.

Model limitation

Quarter turns compress continuous rotation and do not show exact local times, seasons, weather or Earth's scale.

Independent evidence

Four ordered rows recording marker position, light state, apparent Sun position and one final because conclusion.

Supported: Complete four rows using picture choices for sunrise, day, sunset and night. Use: The place is ___ because it faces ___. The Sun appears ___.

Standard: Record four quarter turns and explain the complete cycle using rotation and apparent movement. Include what stayed fixed, what moved and two linked observations.

Stretch: Use the four rows to show how west-to-east rotation explains the pattern. Compare this explanation with the cloud and daily-Sun-orbit ideas without claiming the model alone decides between every possible sky model. Cite exact observations, state the apparent east-to-west direction and include one model limitation.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_G_W8B_Day_And_Night_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.